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RESEARCH ARTICLE

MemesViTa: A Novel Multimodal Fusion Technique for Troll Memes Identification

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ABSTRACT The proliferation of troll memes on social media platforms has become a pressing issue due to their potential to spread misinformation and incite conflict. Detecting troll memes is complex due to the intricate interplay between visual and textual elements. In this research, we propose MemesViTa, a novel multimodal fusion model that combines Vision Transformer (ViT) for image processing and DeBERTa for textual analysis, specifically designed to detect troll memes with high accuracy. This research uses a public dataset of 4,368 training image-text and 1092 testing image-text. The primary goal of this study is to identify the most effective approach for troll meme detection by evaluating the performance of different multimodal models. We comprehensively assess multimodal deep learning approaches for troll meme detection, leveraging visual, textual, and Large Language Models in zero-shot and few-shot scenarios. Our study integrates models including BERT, GPT-2, GPT-4, VGG16, ResNet50, and ViT to analyze their performance. Experimental results indicate that the proposed ViT+DeBERTa model, named MemesViTa, achieves superior results, with 94.287% accuracy and 95.82% F1 score, representing a 41.3% improvement over the best visual model of VGG16 and a 48.1% improvement over the best textual model of CNN. Standalone LLMs like GPT-4, even in few-shot scenarios, show promising yet inferior results, with GPT-4 in a 50-shot scenario reaching an accuracy of 80.00% and an F1 score of 37.00%. Multimodal combinations, including BERT+VGG16 and BERT+VGG16+CLIP, highlight the importance of integrating multiple data modalities. Our findings suggest that combining visual and textual data and the strategic use of LLMs can significantly improve the robustness and accuracy of troll meme detection systems. In the future, we intend to enhance our dataset by incorporating a broader spectrum of troll memes. This will help our model to work better across different social media platforms and contexts.

INDEX TERMS Deep learning, troll memes, text, multimodal, GPT, large language model.

I. INTRODUCTION

The rise of troll memes on social media has raised significant concerns due to their ability to spread misinformation and hate speech. Troll memes are often crafted to provoke emotional responses, incite conflict, and disseminate false information, posing a challenge for content moderation sys-

tems. These memes typically combine provocative imagery with inflammatory text, creating a potent mix that can easily capture attention and influence public opinion. The proliferation of such content necessitates robust detection mechanisms to maintain healthy and safe online environments.

Recent Deep-learning advancements have unlocked new possibilities for troll meme detection. Multimodal DL techniques, which integrate visual and textual information, offer a promising solution. By combining insights from different

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modalities, these models can achieve a more comprehensive understanding of the content and context of memes. The advent of Large Language Models (LLMs) such as BERT, GPT-2, and GPT-4 has revolutionized Natural Language Processing (NLP), providing powerful tools for understanding and generating human language [1], [2], [3].

Multimodal learning techniques are effective in various domains, including image captioning, visual question answering, and emotion recognition. These techniques leverage the complementary strengths of different data types, enabling a more nuanced and accurate analysis. For instance, models that combine convolutional neural networks (CNNs) for image analysis with recurrent neural networks (RNNs) or transformers for text analysis have improved performance in tasks requiring an understanding of visual and textual elements. In the context of troll meme detection, such multimodal models can capture the nuanced interactions between images and text, providing a more holistic view of the content.

The effectiveness of these multimodal models often depends on the quality and diversity of the training data. Large datasets encompassing many meme styles and themes are crucial for training models that generalize well to new, unseen content. Additionally, using advanced pre-training techniques, such as those employed in BERT and GPT, can further enhance the model's ability to understand and generate contextually relevant content. These pre-trained models can then be fine-tuned on specific datasets to adapt to the unique characteristics of troll memes.

Zero-shot learning refers to recognizing and classifying memes it has never seen before, based solely on its understanding of the underlying concepts [4]. In contrast, few-shot learning involves training the model with a few labeled examples, enabling it to quickly adapt to new tasks with minimal data. These learning paradigms are particularly relevant in troll meme detection, where new and varied types of content constantly emerge.

Previous research has introduced various challenges and methodologies, such as the Hateful Memes Challenge, which focuses on detecting hate speech in memes by providing valuable datasets and baselines [5]. Studies have also explored integrating textual, visual, and spatial information for social media data analysis, offering insights into multimodal approaches relevant to troll meme detection [6]. The creation and evaluation of multimodal representations for hate speech detection have been discussed, laying the foundation for our methodologies [7]. Specifically, multimodal DL approaches for troll meme detection have highlighted pertinent challenges and methodologies [8].

Traditional approaches to meme detection have primarily relied on either visual or textual analysis, which limits their effectiveness. Visual models analyze the image content, focusing on features such as color, texture, and shapes [9], [10]. In contrast, textual models concentrate on the captions or text within the memes, examining aspects like sentiment, semantics, and syntax [11]. These multimodal approaches

fail to capture the complex interplay between visual and textual elements that characterize troll memes. For instance, the visual component may convey a sarcastic tone further emphasized by the accompanying text [12].

We explore the performance of GPT-2 and GPT-4 models in zero-shot and few-shot learning scenarios, emphasizing their potential for real-time content moderation. Zero-shot learning enables the classification of unseen content, while few-shot learning allows rapid adaptation to new tasks with minimal labeled data. This study also evaluates the effectiveness of multimodal deep-learning techniques for troll meme detection. We investigate various model architectures that integrate visual models (such as VGG16 [9], ResNet50 [10] and ViT [13]), textual models (such as CNN and LSTM [11]) and LLMs (such as BERT [14], GPT-2 [2] and GPT-4 [3]). Our evaluation encompasses zero-shot and few-shot learning scenarios to assess different models and LLMs' ability to generalize from limited examples. The need for a new approach arises from the limitations of existing multimodal methods, which often fail to adequately capture the complex interactions between visual and textual elements in troll memes, leading to suboptimal detection performance. Our proposed MemesViTa method addresses these shortcomings by leveraging more sophisticated fusion strategies and state-of-the-art models to improve detection accuracy and robustness. Our experiments are designed to determine which visual, textual, and LLM combinations yield the best results regarding accuracy, precision, recall, and F1 score. Our study makes several key contributions as follows:

- Propose a novel MemesViTa model that attains state-of-the-art performance in troll meme detection, with an accuracy of 94.287% and an F1 score of 95.82%.
- Explore the performance of GPT-2 and GPT-4 models in various training scenarios to highlight their ability for real-time content moderation.
- Conduct a detailed performance comparison of visual, textual, and multimodal models to provide insights into their strengths and weaknesses in detecting troll memes.
- Evaluate models in zero-shot and few-shot scenarios to emphasize the practical applicability of these methods in environments where labeled data is scarce or constantly evolving.
- Integrate text and image data to demonstrate the enhanced efficacy of multimodal models in identifying troll content. Our novel techniques leverage state-of-the-art models such as ViT and DeBERTa.

The organization of this paper is as follows: Section II dives into a review of recent studies relevant to the classification of memes data. In Section III, we provide a detailed description of the proposed framework, explaining its algorithmic foundations. Sections IV offer an extensive analysis of experimental comparisons between our work and other prevalent methods, including a series of ablation studies. Finally, Section V summarizes the findings of this research study.

II. LITERATURE REVIEW

The detection and correction of hate speech in multimodal memes [15], which combine visual and textual elements, presents a unique and challenging task for researchers. With the rise of LLMs and visual language models (VLMs), there is a growing interest in leveraging these technologies to address the problem of hateful content in memes. This literature review synthesizes the essential findings and methodologies from recent research efforts.

A. LARGE VISUAL LANGUAGE MODELS (VLMs)

Recent research investigates the capability of large VLMs like LLaVA and Flamingo in detecting and correcting hateful memes. This study demonstrates the effectiveness of pre-trained models in performing zero-shot prompting for hate speech detection and correction tasks [16]. The strengths and weaknesses of these models in dealing with multimodal content are highlighted, emphasizing their potential applications on social media platforms. Ethical considerations and possible biases of these models are also discussed, stressing the importance of continuous refinement and monitoring to ensure fair and accurate outcomes. Our approach enhances the integration of textual and visual elements, aiming to improve accuracy and reduce biases in meme detection through advanced multimodal fusion techniques. We address the limitations identified in previous studies by implementing a robust multimodal fusion mechanism that improves alignment between image and text data, resulting in better detection performance. Additionally, our methodology incorporates a more extensive dataset, leveraging diverse and representative samples to train the model, which helps mitigate bias and enhance the generalizability of the results. Focusing on the nuanced interplay between visual and textual cues, our approach provides a more comprehensive framework for understanding and detecting hateful content.

B. MULTIMODAL LEARNING APPROACHES

In this study [17], a novel approach incorporating image captioning into the hateful memes detection process is proposed. The model addresses the challenge of weakly aligned or irrelevant visual and textual information in memes. The experimental results show promising performance on the Hateful Memes Detection Challenge, demonstrating the model's effectiveness in understanding and reasoning over multiple modalities. Furthermore, the scalability of this approach is highlighted, along with its potential to be adapted to various other applications requiring multimodal understanding, such as content moderation and automated annotation systems.

This work [18] introduced a multimodal few-shot learning approach that leverages a frozen language model to generate image captions. This method lets the model learn new tasks with few examples, ideal for visual question-answering and object recognition. The study underscores the advantages of using frozen language models in reducing computational

costs and training time while achieving high accuracy. This approach has also been explored with other AI systems to enhance its functionality and application scope. Our work extends these findings by implementing a more robust multimodal fusion mechanism that improves the alignment between image and text data, resulting in better detection performance. We also incorporate dynamic learning techniques that reduce computational overhead while maintaining high accuracy in detecting complex meme structures.

C. EXPLAINABILITY AND ERROR ANALYSIS

The decision-making process of multimodal hateful meme detection models is examined in [15], focusing on whether these models can accurately capture derogatory references and perform visual-text slur grounding. The error analysis reveals biases in the models, resulting in false-positive predictions, highlighting the need for improved interpretability in VLMs. Transparency in AI systems is emphasized to build trust and ensure that these models can be effectively used in sensitive applications such as content moderation on social media. A novel framework is introduced to identify and mitigate biases, addressing the shortcomings highlighted in this study.

An explainable approach using multimodal debate between LLMs to detect harmful memes is proposed in [19]. The model performs dialectical reasoning over implicit harmful patterns by eliciting contradictory arguments from LLMs and employing a debate judge for inference. This method shows superior performance and explanation capabilities on multiple public meme datasets. The study suggests that such debate-based frameworks could enhance the robustness and interpretability of AI systems, making them more reliable for real-world deployment.

D. SAFETY AND BENCHMARKING

A comprehensive benchmark for assessing the safety and effectiveness of Large Multimodal Models (LMMs) in detecting social abuse in memes is introduced in [20]. The study reveals deficiencies in current models' safety awareness, underscoring the critical need for advancements to ensure safe artificial intelligence. The benchmark includes diverse, challenging scenarios to test the models' ability to handle nuanced and potentially harmful content. The argument is made that continuous benchmarking and rigorous Testing are essential for developing AI systems that are both effective and safe in identifying and mitigating social abuse. Our framework includes enhanced safety protocols and thorough Testing to ensure robust, ethical troll meme detection, reducing false positives and negatives for improved reliability.

E. PROMPT-BASED MODELS

This study [21] presents PromptHate, a prompt-based model that leverages the implicit knowledge in pre-trained language models for hateful meme classification. Their approach

performs well with simple prompts and in-context examples, outperforming state-of-the-art baselines on publicly available datasets. The authors emphasize the flexibility and scalability of prompt-based models, highlighting their potential to adapt to various contexts and tasks with minimal retraining.

F. MULTIMODAL NEURAL NETWORKS

This research [22] proposes a multimodal deep neural network for hateful meme detection. Their approach integrates textual and visual branches in a final classification step, tested on the Memotion 7k dataset. The results demonstrate the model's capability to handle the multimodal nature of Internet memes, showing significant improvements over unimodal models. The study suggests that the synergy between visual and textual information is crucial for accurately understanding and detecting hate speech in memes. Our work enhances multimodal neural networks with advanced fusion techniques, improving integration and detection performance.

G. FRAMEWORKS FOR HARMFUL MEME DETECTION

This work [23] introduces MOMENTA, a multimodal framework that analyzes both local and global perspectives of memes to detect harmful content and identify targeted entities. Their framework shows superior interpretability and generalizability, outperforming existing approaches on the HarMeme dataset. The authors argue that considering detailed and overarching contexts is essential for robust harmful content detection. They propose extending their framework to other types of multimodal content beyond memes.

H. RESEARCH GAP AND QUESTIONS

The reviewed literature highlights significant advancements in detecting and correcting hate speech in multimodal memes using large visual language models and multimodal learning approaches. Future research should focus on enhancing model interpretability, addressing biases, and improving the safety awareness of multimodal models to effectively mitigate the spread of harmful content on social media platforms. There is also a call for more comprehensive and diverse datasets to train and evaluate these models better, ensuring their robustness and applicability in real-world scenarios. Our novel multimodal fusion approach enhances interpretability, bias mitigation, and safety, providing a more accurate and ethical solution for troll meme detection. Table 1 shows a thorough analysis of the different utilized methodologies, the dataset, the techniques implemented, and the outcomes associated with the existing memes detecting methods.

III. METHODOLOGY

This section outlines our study's various models and approaches for detecting troll memes. We explore visual models such as VGG16, ResNet50, and VGG19, textual models including BERT and CNNs, and multimodal learning

approaches that combine visual and textual data. Additionally, we evaluate the performance of LLMs like GPT-2 and GPT-4 in zero-shot and few-shot learning scenarios. Our methodology involves detailed image and text data preprocessing, integrating different model architectures, and a rigorous training procedure with specific hyperparameters to ensure robust performance metrics. Figure 1 depicts the workflow of the proposed method for this study.

A. VISUAL MODELS

1) VGG16 MODEL

VGG16 is a CNN with 16 layers, including 13 convolutional layers and three fully connected layers, characterized by small 3×3 filters that capture detailed spatial hierarchies in images [9], [26]. VGG16 was chosen for its ability to capture fine-grained visual features using these small convolutional filters. Its deep architecture excels in tasks requiring identifying intricate patterns in images, making it well-suited for detecting the nuanced visual cues in troll memes.

While VGG16 excels in image classification, it cannot process textual data, which is crucial for interpreting troll memes that rely on text to convey their message. The convolutional operation in VGG16 is represented as:

$$Z_l = \sigma(W_l * Z_{l-1} + b_l) \quad (1)$$

where Z_l is the activation at layer l , W_l are the weights, b_l are the biases and σ is the activation function [27]. Integrating VGG16 with models capable of textual analysis is essential for comprehensive meme analysis. Future work should focus on creating hybrid models that combine visual and textual data to improve troll meme detection.

2) ResNet50 MODEL

ResNet50, or Residual Networks, employs a residual learning framework with 50 layers and shortcut connections to tackle the vanishing gradient issue [10]. ResNet50 was chosen because it introduces residual connections that help solve the vanishing gradient problem in deep networks. This allows it to capture more profound and more complex visual features. Its effectiveness in large-scale image classification tasks makes it an ideal candidate for detecting visually complex memes, especially when combined with textual information. ResNet introduces residual blocks that allow the model to learn identity mappings [28], mitigating vanishing gradients. Each residual block is represented as:

$$y = F(x, \{W_i\}) + x \quad (2)$$

where x is the input, F is the residual mapping and $\{W_i\}$ are the weights [10].

3) VGG19 MODEL

VGG19 is an extended version of VGG16, consisting of 19 layers, including 16 convolutional layers and three fully connected layers. Like VGG16, VGG19 employs

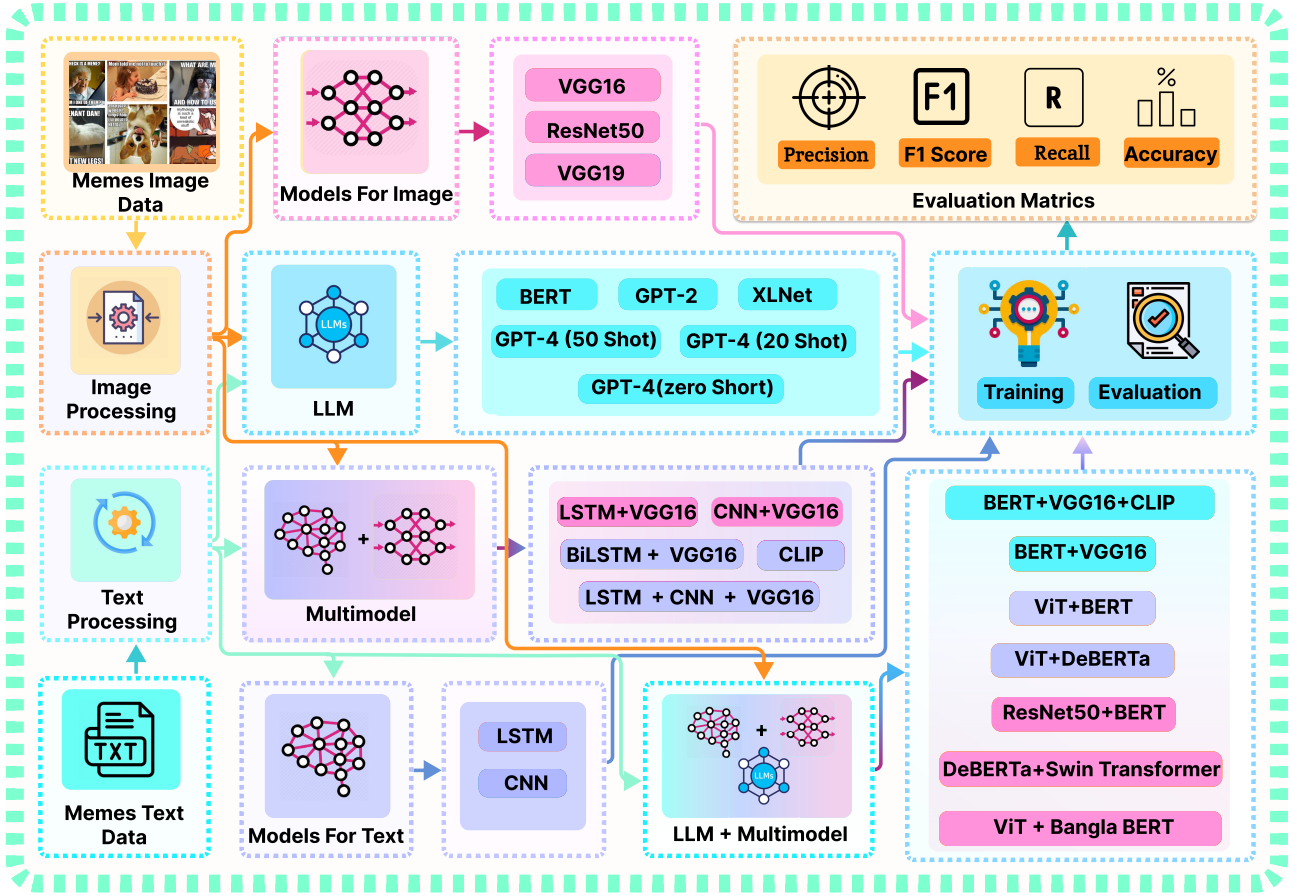


FIGURE 1. The workflow of proposed multimodal DL approaches for troll meme detection method.

small 3×3 filters throughout the network to capture detailed spatial hierarchies in images. The increased depth allows VGG19 to learn more complex features from the pictures.

The convolutional operation in VGG19 can be represented as:

$$Z_l = \sigma(W_l * Z_{l-1} + b_l) \quad (3)$$

where Z_l is the activation at layer l , W_l are the weights, b_l are the biases and σ is the activation function.

B. TEXTUAL MODELS

1) BERT MODEL

BERT is a transformer-based model developed to pre-train deep bidirectional representations by considering left and right contexts in all layers. BERT was chosen for this task due to its efficient understanding of text context. Troll memes often involve sarcastic or misleading text, and BERT's capacity to model long-range dependencies in language enables it to detect these subtle textual nuances, making it ideal for identifying the textual components of troll memes.

BERT uses a transformer architecture with multiple layers of self-attention. The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

where Q (queries), K (keys) and V are matrices derived from the input and d_k is the dimensionality of the keys.

2) DistilBERT MODEL

DistilBERT (Distilled BERT) is a compact, efficient version of BERT created by Hugging Face through model distillation. This technique trains a smaller model to imitate a larger one, resulting in DistilBERT being 60% faster and 40% smaller than BERT while retaining 97% of its language understanding capabilities. DistilBERT was chosen for troll meme detection due to its efficiency. It is highly suitable for real-time content moderation systems where speed and resource usage are critical without significantly compromising language understanding. DistilBERT reduces BERT's layers from 12 to 6, using the same hidden size and number of attention heads, maintaining robust performance in detecting nuanced textual elements.

DeBERTa improves upon BERT by disentangling content and position information in language, making it more precise

TABLE 1. Summary of recent studies on multimodal DL approaches for troll memes detection.

Ref.	Year	Approach & Methodology	Dataset Characteristics	Classification Strategy	Performance Results
[16]	2023	Detecting Hate Speech in Multimodal Memes with LLMs	Facebook Hateful Memes dataset	pre-trained language models (PLMs)	AUROC of 90.96% and 84.47%
[17]	2020	Multimodal Learning for Hateful Memes Detection	Hateful Memes Challenge dataset	Image captioner, object detector and triplet-relation network	AUROC score of 75.44
[19]	2024	Towards Explainable Harmful Meme Detection through Multimodal Debate between LLMs	Harm-C, Harm-P and Facebook Hateful Memes datasets	Multimodal debate mechanism with LLMs	Accuracy of 87.00%, macro-averaged F1 score of 86.41% on Harm-C; 90.73% and 90.72% on Harm-P; 75.60% and 75.39% on FHM
[20]	2024	Safety Insights to Large Multimodal Models	GOAT-Bench dataset	GPT-4V, CogVLM and other state-of-the-art LLMs	Accuracy of 72.17%, macro-averaged F1 score of 70.29%
[21]	2021	Prompting for Multimodal Hateful Memes Classification	Facebook Hateful Memes dataset, Harmful Memes dataset	PromptHate framework using RoBERTa language model	AUC of 90.96 and accuracy of 84.47 on HarM dataset; AUC of 81.45 and accuracy of 72.98 on FHM dataset
[18]	2021	Multimodal Few-Shot Learning with Frozen Language Models	Conceptual Captions dataset, VQAv2, OKVQA, miniImageNet, Visual Genome	Frozen method integrating images and text embeddings	VQAv2 zero-shot accuracy of 29.5%, 38.2% with four shots; OKVQA zero-shot accuracy of 5.9%, 12.6% with four shots
[22]	2021	Hateful Memes Detection with Multimodal Deep Neural Networks	Memotion 7k dataset	Parallel processing branches for text and visual data, combined in final classification	Macro F1 score of 0.2498, 8.24% improvement over baseline DNN architecture
[24]	2023	Multimodal Religiously Hateful Social Media Memes Classification based on Textual and Image Data	Facebook Hateful Memes dataset	VisualBERT	AUROC is 0.7899 and Accuracy is 0.7060
[23]	2021	MOMENTA: A Multimodal Framework for Detecting Harmful Memes and Their Targets	Harm-C and Harm-P datasets	CLIP for image and text embedding, VGG-19 and DistilBERT for object proposals	Accuracy on Harm-C for harmful memes detection is 83.82%, target identification is 77.95%
[25]	2020	Detecting Hate Speech in Memes Using Multimodal DL Approaches	Hateful Memes Challenge	VisualBERT with ensemble learning	AUROC score of 0.811, with an accuracy of 0.765

in understanding textual dependencies. Its advanced attention mechanism is crucial in detecting the complex textual content often found in troll memes.

3) XLNet MODEL

XLNet is a generalized autoregressive pretraining method for language understanding developed by Google and Carnegie Mellon University. XLNet uses a transformer-XL architecture, which includes a segment-level recurrence mechanism and a novel relative positional encoding scheme, allowing it to model longer-term dependencies more effectively than traditional transformers.

4) MEMES BERT

Memes BERT is a language model specifically tailored for the language based on the BERT framework. It has been trained on an extensive collection of Memes texts, enabling it to effectively handle NLP tasks like text classification, sentiment analysis, and language translation.

5) CNNs FOR TEXT MODEL

CNNs, while traditionally used for image processing, have also been adapted for text classification [29]. In this context, convolutional layers apply filters over word embeddings to capture local patterns and n-grams in the text. The convolution operation for text can be expressed as:

$$c_i = f(W \cdot x_{i:i+h-1} + b) \quad (5)$$

where $x_{i:i+h-1}$ is the concatenation of word embeddings, W is the filter, b is the bias and f is a non-linear activation function.

6) LSTM MODEL

Long Short-Term Memory (LSTM) networks are a type of RNN that can learn long-term dependencies, which is essential for understanding the context in sequential data such as text [30]. LSTMs address the vanishing gradient problem by incorporating memory cells that can retain information over extended periods.

The following equations describe the operations in an LSTM cell:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (6)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (7)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (8)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (9)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (10)$$

$$h_t = o_t * \tanh(C_t) \quad (11)$$

where f_t is the forget gate, i_t is the input gate, $\tilde{C}_t$ is the candidate cell state, C_t is the cell state, o_t is the output gate, h_t is the hidden state, x_t is the input at time step t and σ and $\tanh$ are the sigmoid and hyperbolic tangent activation functions, respectively.

C. MULTIMODAL MODELS

Multimodal Learning Approaches: Multimodal learning combines visual and textual information to improve meme detection accuracy. This approach leverages the complementary strengths of different data types, enabling a more comprehensive analysis of memes. Advanced language models, such as BERT and CLIP (Contrastive Language-Image Pretraining), align images and text in a shared embedding space.

CLIP works by jointly training an image encoder and a text encoder to predict which images are paired with which texts in a dataset. The objective function for CLIP is to maximize the cosine similarity between the correct image-text pairs while minimizing it for incorrect pairs:

$$L = -\log \frac{\exp(\text{sim}(I, T^+))}{\sum_T \exp(\text{sim}(I, T))} \quad (12)$$

where $\text{sim}(I, T)$ is the cosine similarity between the image I and text T and T^+ represents the correct text paired with image I .

For example, a multimodal model might combine the outputs of a CNN for image data and a BERT model for text data:

$$h_{\text{multimodal}} = \text{concat}(h_{\text{visual}}, h_{\text{textual}}) \quad (13)$$

where h_{visual} and h_{textual} are the hidden representations from the visual and textual models, respectively.

1) LARGE LANGUAGE MODELS (LLMs)

GPT-2 and GPT-4 Model: GPT-2 and GPT-4 were selected for their generative solid capabilities, especially in zero-shot and few-shot learning scenarios. In a rapidly evolving domain like troll meme detection, where new types of content frequently emerge, GPT models offer the flexibility to classify content with minimal labeled data, making them highly adaptable. Their ability to generalize from limited examples is beneficial in detecting novel forms of troll memes with minimal supervision.

GPT-2 and GPT-4 are autoregressive language models that use a transformer architecture to generate human-like text. These models are pre-trained on diverse text data and can generate coherent and contextually relevant text [31], making them valuable for analyzing meme captions.

The autoregressive generation process [32] can be defined as:

$$P(x) = \prod_{t=1}^T P(x_t | x_{<t}) \quad (14)$$

where x_t is the token at time step t and $x_{<t}$ are the preceding tokens. GPT-4 can generate text based on minimal context in zero-shot and few-shot learning scenarios. The performance in these scenarios can be quantified as follows:

- **Zero-Shot:** Zero-shot learning refers to the model's ability to generate or predict the specific task without

prior examples or context.

$$P(x) = \prod_{t=1}^T P(x_t | \emptyset) \quad (15)$$

- **20-Shot:** In 20-shot learning, the model is given 20 examples or pieces of context to learn from before generating or predicting.

$$P(x) = \prod_{t=1}^T P(x_t | x_{<t}^{20}) \quad (16)$$

- **50-Shot:** In 50-shot learning, the model is given 50 examples or contexts to learn from before generating or predicting.

$$P(x) = \prod_{t=1}^T P(x_t | x_{<t}^{50}) \quad (17)$$

D. DATASET OVERVIEW

The dataset used in this study comprises troll and non-troll memes containing 3,264 images. These images were split into three subsets: 80% for training, 10% for validation, and 10% for testing [33]. Specifically, 2,613 images were used for training, 326 for validation, and 815 for Testing. This division ensured an effective distribution for model training, hyperparameter tuning, and evaluation of unseen data.

We also employed k-fold cross-validation (k=5) to enhance the robustness of the model. The dataset was randomly split into training and validation subsets for each fold. The average performance across all folds was recorded.

Data imbalance between troll and non-troll classes was addressed through oversampling and augmentation techniques such as image flipping, rotation, and cropping. This approach helped create a balanced training dataset and improved the model's generalization ability.

Regarding pretraining, we utilized Vision Transformer (ViT) and DeBERTa models pre-trained on large datasets (ImageNet for ViT and BookCorpus for DeBERTa) to leverage transfer learning. These models were subsequently fine-tuned on the troll meme dataset for more precise detection of troll content. The dataset distribution is presented in Table 2, with a bar plot illustrating the proportion of troll and non-troll labels, as shown in Figure 2.

TABLE 2. Dataset distribution.

Label	Troll	Not-troll	Total
Train	2184	2184	4368
Test	546	546	1092

As shown in Figure 3, a sample troll meme illustrates the typical combination of provocative imagery and text that these memes use to incite reactions.

E. PREPROCESSING

Image Preprocessing: The image data used in this study were preprocessed to ensure consistency and compatibility with the

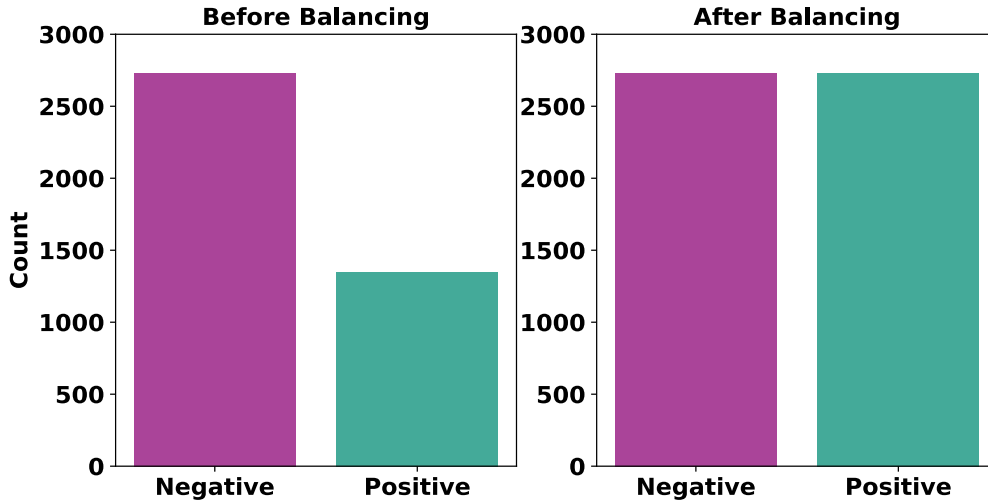


FIGURE 2. Distribution of memes before and after balancing the dataset.

Vision Transformer (ViT) model. All images were resized to a standard resolution of 224×224 pixels, which is required for ViT input. After resizing, each image was normalized using the standard ImageNet mean and standard deviation values (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). This normalization process transforms each pixel value x_i of the image according to the equation:

$$x'_i = \frac{x_i - \mu}{\sigma}$$

where x_i is the original pixel value, μ is the mean of the dataset, and σ is the standard deviation.

Text Preprocessing: The text data consisted of captions associated with the memes and were preprocessed using the DeBERTa tokenizer. The text preprocessing pipeline involved several vital steps to prepare the data for input into the DeBERTa model:

1) TOKENIZATION

Each sentence was tokenized into subword tokens using the DeBERTa tokenizer. This process converts the input text into a sequence of tokens that the transformer model can process. Unique tokens, such as [CLS] and [SEP], were added to mark the beginning and end of each sentence, respectively.

2) PADDING AND TRUNCATION

To ensure that all text inputs have the same length, the tokenized sequences were padded or truncated to a maximum sequence length of 128 tokens. This ensures uniformity and allows for efficient batch processing.

3) NORMALIZATION

The text was lowercase, and unnecessary punctuation and non-alphanumeric symbols were removed. This step reduces noise in the dataset, improving the model's ability to generalize.

4) STOP WORDS AND LEMMATIZATION

Common stop words were removed, and lemmatization was applied to reduce words to their base form. These steps were taken to simplify the text data and focus on meaningful content. The input to the DeBERTa model is therefore described as:

$$\text{Input} = (\text{input_ids}, \text{attention_mask})$$

where input_ids corresponds to the tokenized words, and attention_mask is a binary mask used by the model to ignore padded tokens during processing.

Model Training and Hyperparameter Selection: All models, including VGG16, ResNet50, BERT, ViT, and DeBERTa, were trained using a standard cross-entropy loss for classification tasks, with the Adam optimizer adjusting the learning rates adaptively. Each model was trained on the troll meme dataset with the same set of hyperparameters, ensuring fair comparisons across models. The hyperparameters include a learning rate of $1e-4$, a batch size of 16, and ten training epochs.

For multimodal models like ViT + DeBERTa (MemesViTa), the training process involved combining visual and textual representations, ensuring that the model learned to capture the relationships between both modalities. A dropout layer (with a probability of 0.3) was used in the multimodal model to reduce the chances of overfitting.

The hyperparameters for all models used in this study are summarized in Table 3. The performance of each model was evaluated using metrics such as accuracy, precision, recall, and F1 score to assess their effectiveness in detecting troll memes.

We employed several strategies to ensure optimal performance and prevent overfitting. The models were trained for ten epochs based on preliminary experiments that showed diminishing returns after this point. A learning rate of $1e-4$ was chosen through a grid search, providing the



FIGURE 3. Sample troll memes demonstrating the integration of provocative imagery with text to elicit reactions.

best convergence without overfitting. To prevent overfitting, we applied data augmentation techniques such as image flipping, rotation, and cropping to create a more diverse training set. Early stopping was also used, halting training if validation loss did not improve after two consecutive epochs (patience = 2). Additionally, a cosine learning rate decay scheduler dynamically adjusted the learning rate during training, enabling faster convergence initially and more refined updates toward the end.

Integrating various models (vision-based, text-based, and multimodal) aimed to enhance the accuracy and robustness of troll meme detection. A relatively small batch size and a moderate learning rate were chosen to balance training speed and stability across all models. The Adam optimizer was selected for its effectiveness in handling sparse gradients and adaptive learning rates, which is beneficial for training deep networks.

F. PROPOSED NOVEL MemesViTa METHOD: ViT-DeBERTa MODEL

This study proposes a novel multimodal approach for troll meme detection that leverages the strengths of Vision Transformer (ViT) and DeBERTa models. These models were chosen because they offer state-of-the-art capabilities in handling visual and textual information, which is crucial for accurately detecting troll memes. This task often requires understanding complex interactions between image and text.

1) VISION TRANSFORMER (ViT)

ViT was chosen over traditional CNNs because it captures global context and long-range dependencies within images. Unlike CNNs focusing on local features, ViT's transformer-based architecture treats an image as a sequence of patches, allowing it to detect subtle visual cues and

broader patterns in troll memes. This is especially important for memes, where the interplay between imagery and text often conveys sarcastic, misleading, or harmful content. ViT's ability to analyze images in a tokenized form makes it more adept at understanding visual relationships that traditional models might miss, resulting in more robust meme detection.

ViT is a transformer-based model designed for image classification tasks. Unlike traditional CNNs, ViT divides an image into a sequence of patches, which are then linearly embedded, combined with position embeddings, and fed into a standard transformer encoder [13]. This approach allows ViT to capture global context and complex patterns within images, making it highly effective for visual data analysis. The ViT model can be mathematically represented as follows:

$$Z_0 = [x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{pos} \quad (18)$$

where x_p^i represents the i -th image patch, E is the linear projection of patches and E_{pos} denotes the positional embedding. The sequence Z_0 is then processed through transformer layers to generate the final visual representation.

2) DeBERTa

DeBERTa was selected over other text models like BERT due to its improved disentangled attention mechanism, better capturing the complex semantics of meme text. Troll memes frequently rely on irony, sarcasm, or coded language, and DeBERTa's ability to model content and positional information separately makes it particularly suited for nuanced textual analysis. This is critical for understanding the intent behind troll memes, as subtle shifts in wording or tone can drastically change their meaning. DeBERTa's advanced attention mechanisms help it detect these changes more accurately than other models.

TABLE 3. Hyperparameter settings for all models.

Component	Details
Vision Models	
VGG16	Pretrained VGG16 (13 conv layers), Batch size: 16, Epochs: 10, Learning Rate: 1e-4, Adam Optimizer, Dropout: No, Early Stopping: Patience 2
ResNet50	Pretrained ResNet50 (50 layers), Batch size: 16, Epochs: 10, Learning Rate: 1e-4, Adam Optimizer, Dropout: No, Early Stopping: Patience 2
ViTModel	ViTModel (google/vit-base-patch16-224-in21k), Batch size: 16, Epochs: 10, Learning Rate: 1e-4, Adam Optimizer, Dropout: No, Early Stopping: Patience 2
Text Models	
BERT	BERTModel (bert-base-uncased), Batch size: 16, Epochs: 10, Learning Rate: 1e-4, Adam Optimizer, Dropout: No, Early Stopping: Patience 2
DeBERTa	DebertaModel (Microsoft/deberta-base), Batch size: 16, Epochs: 10, Learning Rate: 1e-4, Adam Optimizer, Dropout: No, Early Stopping: Patience 2
Multimodal Model	
MemesViTa (ViT+DeBERTa)	Linear layer (input: 768+768, output: 2), Batch size: 16, Epochs: 10, Learning Rate: 1e-4, Adam Optimizer, Dropout: 0.3, Early Stopping: Patience 2

3) INTEGRATION OF ViT AND DeBERTa

ViT and DeBERTa combine to provide a comprehensive multimodal understanding of memes by integrating visual and textual data. While ViT captures the broader visual context, DeBERTa dissects the intricacies of meme text, ensuring that the model recognizes visual patterns and textual semantics. This integration is essential for detecting troll memes, often using text to reinforce or mask the visual message. By combining these two models, MemesViTa leverages the complementary strengths of both modalities, resulting in significant improvements in detection accuracy. The architecture is shown in Figure 4.

The multimodal representation $h_{\text{multimodal}}$ is obtained by:

$$h_{\text{multimodal}} = \text{concat}(h_{\text{visual}}, h_{\text{textual}}) \quad (19)$$

where h_{visual} is the output from ViT and h_{textual} is the output from DeBERTa. The concatenated representation is then passed through a fully connected layer to generate the final prediction.

Traditional models like CNNs and RNNs were considered but found lacking in their ability to process the multimodal complexity of troll memes. CNNs, for example, excel at local feature detection but struggle with capturing the global image context necessary to understand visual humor or intent. RNNs, on the other hand, often fail to handle the subtle textual dependencies [34] in meme text. ViT and DeBERTa outperform these models due to their transformer-based architectures, which are specifically designed to capture long-range dependencies in visual and textual data. Moreover, models like ViT are more adaptable to dynamic meme content, where the relationship between image and text constantly evolves.

IV. EXPERIMENTAL RESULTS

The following section details the experimental setup and the results of evaluating the proposed MemesViTa multimodal model for troll meme detection. The experiments were conducted on a comprehensive multimodal meme dataset encompassing images and text captions. The model's performance was systematically evaluated using a variety of

metrics, including accuracy, precision, recall, and F1 score, ensuring a thorough assessment of its capabilities.

Algorithm 1 Proposed Novel MemesViTa Model

Input: Multimodal Meme Dataset

Output: Troll Meme Detection | Troll or Non-Troll ViT + BERT Multimodal Model

Initialize ViT and BERT models

Split data a. **Training set (TrS)** b. **Testing set (TeS)**

Train ViT model using the training set (TrS)

a. $\text{ViT}_{\text{Model}} \leftarrow \text{Train}_{\text{ViT}}(\text{TrS})$

Train BERT model using the training set (TrS)

a. $\text{BERT}_{\text{Model}} \leftarrow \text{Train}_{\text{BERT}}(\text{TrS})$ i in the testing set (TeS)

Predict using ViT model a. $\text{ViT}_p \leftarrow \text{Predict}_{\text{ViT}}(i)$

Predict using BERT model a. $\text{BERT}_p \leftarrow \text{Predict}_{\text{BERT}}(i)$

Combine predictions to form the final prediction

a. $\text{Final}_{\text{pred}} \leftarrow \text{MajorityVote}(\text{ViT}_p, \text{BERT}_p)$ majority vote prediction is **Troll or Non-Troll**

a. Output the final prediction: **Troll or Non-Troll**

A. EVALUATION METRICS

We used the following evaluation metrics to assess the models' performance. The abbreviations TP, FP, FN, and TN are True Positive, False Positive, False Negative, and True Negative, respectively.

Accuracy: The proportion of correctly classified instances among the total instances. It provides an overall measure of how well the model performs across all classes. The formula for accuracy is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (20)$$

Precision: The proportion of true positive results among all positive results predicted by the model. The formula for precision is:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (21)$$

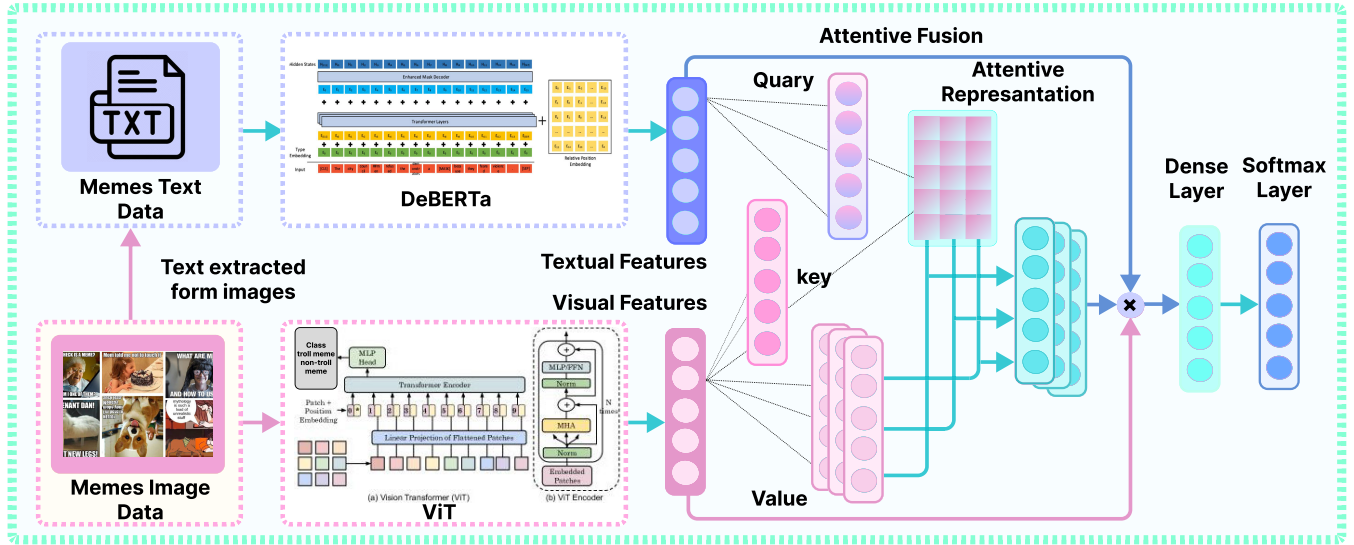


FIGURE 4. Proposed novel MemesViTa model architecture.

Recall: The recall is the ratio of correctly identified positive instances to all actual positive cases as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (22)$$

F1 Score: The harmonic mean of precision and recall, providing a single metric that balances both concerns. The formula for F1 Score is:

$$F1 \text{ Score} = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Precision} + \text{Recall}} \quad (23)$$

B. RESULTS FOR VISUAL MODELS

The performance of the visual models is summarized in Table 4. Among the visual models evaluated, VGG16 achieved the highest accuracy of 66.74%. This indicates that VGG16 was the most effective at identifying visual patterns and features relevant to troll meme detection among the visual models tested. The precision, recall, and F1 score metrics further illustrate VGG16's capability in distinguishing between troll and non-troll memes. ResNet50 and VGG19, while slightly behind in performance, also demonstrated significant potential in visual meme classification. ResNet50 achieved an accuracy of 66.17%, showcasing its strength in handling deep neural networks with skip connections. VGG19, despite having a more complex architecture, achieved an accuracy of 63.09%, suggesting that deeper networks do not always guarantee better performance for this task in Figure 5. The confusion matrices in Figure 5 provide a detailed view of the classification results, highlighting the areas where each model excels or needs improvement.

C. RESULTS FOR TEXTUAL MODELS

The results for the textual models are presented in Table 4. The CNN model achieved the highest accuracy of 63.66%, demonstrating its capability in handling text classification

tasks, such as analyzing meme captions for troll content. This performance underscores the effectiveness of CNNs in capturing spatial features within textual data, which are crucial for distinguishing between troll and non-troll memes. The LSTM+CNN model also performed well, achieving an accuracy of 62.06%, highlighting the advantage of combining sequential and spatial processing capabilities. This combination allows the model to understand the text's context and structure better, enhancing its classification performance. Additionally, despite being a powerful language model, BERT achieved an accuracy of 63.20%, suggesting that the task may benefit more from models that can capture both sequential and spatial features.

D. RESULTS FOR MULTIMODAL MODELS

Multimodal models, which combine visual and textual features, were evaluated to assess their ability to leverage both data types for improved performance. As shown in Table 6, the combination of ViT and DeBERTa outperformed other multimodal models, achieving the highest accuracy of 94.287%. This indicates the advantage of integrating advanced visual and textual models for troll meme detection. The superior performance of the MemesViTa model can be attributed to ViT's ability to capture intricate visual features and DeBERTa's capability to understand complex textual semantics. This combination allows for a more comprehensive analysis of visual and textual elements within memes, leading to more accurate classification results. Other multimodal models, such as ResNet50+BERT and BERT+VGG16+CLIP, also showed significant improvement over individual modalities, demonstrating the benefits of a multimodal approach. The confusion matrices in Figure 6 provide a detailed view of the classification outcomes, highlighting the strengths and weaknesses of each multimodal configuration.



FIGURE 5. Visual-textual multimodal confusion matrix.

E. ZERO-SHOT AND FEW-SHOT TECHNIQUES

In testing GPT-2 and GPT-4, we evaluated the ability of LLMs to perform in zero-shot and few-shot learning scenarios. The results, shown in Table 5, highlight the performance of these models with minimal training examples. GPT-4 demonstrated substantial potential, achieving an accuracy of 80% in a 50-shot scenario, indicating its ability to generalize from limited data and adapt to troll meme detection effectively. This performance underscores the flexibility and adaptability of GPT-4 in understanding and classifying complex meme content with minimal training data. Additionally, the model's precision and recall scores improved significantly as the number of shots increased, showcasing its learning capabilities even with small datasets. GPT-4's performance exhibits a clear upward trajectory as the number of shots increases from 20 to 50, with the F1 score improving from 24.4% in the 20-shot scenario to 37.00% in the 50-shot scenario. This suggests that as GPT-4 is exposed to more task-specific examples, it adapts better, demonstrating its ability to generalize with minimal data. However, specific characteristics of troll memes, especially those with highly contextual or subtle humor, present a challenge in zero-shot settings, where the model lacks prior examples. In contrast, GPT-4 excels in few-shot scenarios, where the additional context provided by the training examples helps the model interpret the nuanced interplay between image and text more effectively. However, text-heavy memes or those with straightforward visual-text alignment tend to be handled

better by GPT-4 in few-shot settings, as the additional context helps the model learn specific patterns of humor or sarcasm. In contrast, memes with complex cultural references or layered sarcasm, which require a deeper understanding of context, remain challenging in zero-shot settings but see marked improvements as the number of shots increases, particularly in the 50-shot scenario.

Figure 7 provides a detailed view of the confusion matrices, illustrating the classification performance across different scenarios.

These experiments and results highlight the effectiveness of multimodal models and the promising potential of few-shot learning with advanced LLMs like GPT-4 for troll meme detection. The integration of visual and textual features, combined with the advanced capabilities of LLMs, offers a robust approach to addressing the challenges posed by troll memes on social media platforms. This methodology demonstrates that powerful LLMs can achieve high accuracy and adapt to complex tasks even with minimal labeled data. They are invaluable tools for content moderation and automated detection systems. This approach is analogous to the ensemble techniques used in medical image classification to enhance diagnostic accuracy [10], [26].

F. EXPERIMENTAL SETUP

The experiments used an experimental framework with libraries such as Sci-kit-learn, PyTorch, TensorFlow, and additional language modeling tools. The programming

TABLE 4. Evaluation results of visual, textual, and multimodal models.

Approach	Classifier	Accuracy	Precision	Recall	F1 Score
Visual	VGG16	0.6674	0.4230	0.4633	0.4421
	ResNet50	0.6617	0.4163	0.4210	0.4104
	VGG19	0.6309	0.2103	0.3333	0.2579
Textual	CNN	0.6366	0.4514	0.4309	0.4291
	LSTM+CNN	0.6206	0.4507	0.4392	0.4411
Multimodal	LSTM+VGG16	0.6183	0.4346	0.4210	0.4239
	CNN+VGG16	0.6023	0.4395	0.4307	0.4336
	LSTM+CNN+VGG16	0.5977	0.4520	0.4490	0.4458
	BiLSTM+VGG16	0.5989	0.4092	0.4232	0.4139
	CLIP	0.7344	0.7457	0.9150	0.8217

TABLE 5. Performance of only LLM models.

Model	Scenario	Accuracy	Precision	Recall	F1 Score
BERT Only	-	0.6320	0.4107	0.3361	0.2656
DistilBERT	-	0.6045	0.3953	0.4204	0.4075
XLNet	-	0.68	0	0	0
GPT-2	Epoch 1	0.7691	0.8333	0.3285	0.4712
	Epoch 2	0.7989	0.8182	0.4599	0.5888
	Epoch 3	0.8229	0.7874	0.5949	0.6778
GPT-4	Zero-Shot	0.6926	0.5556	0.0185	0.0358
	20-Shot	0.7500	0.6500	0.1500	0.2400
	50-Shot	0.8000	0.7000	0.2500	0.3700

TABLE 6. Multimodal models with LLMs.

Model	Accuracy	Precision	Recall	F1 Score
BERT+VGG16	0.6617	0.4183	0.4595	0.3826
BERT+VGG16+CLIP	0.6971	0.4502	0.4547	0.3160
ResNet50+BERT	0.7168	0.7367	0.8504	0.7895
ViT+BERT	0.9329	0.9407	0.9229	0.9462
MemesViTa	0.94287	0.9525	0.9642	0.9582
DeBERTa+Swin Transformer	0.9314	0.9398	0.9533	0.9465
ViT+Bangla BERT	0.93004	0.9360	0.9491	0.9421

language selected for these implementations was Python 3.10. The testing environment for these experiments was an ASUS ROG Zephyrus G16 GA605WI-XB96, equipped with Windows 11 Pro, an NVIDIA® GeForce RTX™ 4070 Laptop GPU, an AMD Ryzen™ AI 9 HX 370 Processor, and 1TB M.2 NVMe™ PCIe® 4.0 SSD storage. This robust hardware setup could train conventional and LLMs, providing substantial computational power for managing complex neural network models and large datasets.

G. COMPUTATIONAL RESOURCES AND MODEL EFFICIENCY

Training time varied across models based on the number of parameters and data processing complexity. For instance, the ViT + DeBERTa model required approximately 4 hours to complete five training epochs on the dataset. In comparison, standalone visual models like VGG16 and ResNet50 took around 2 hours each for the same number of epochs.

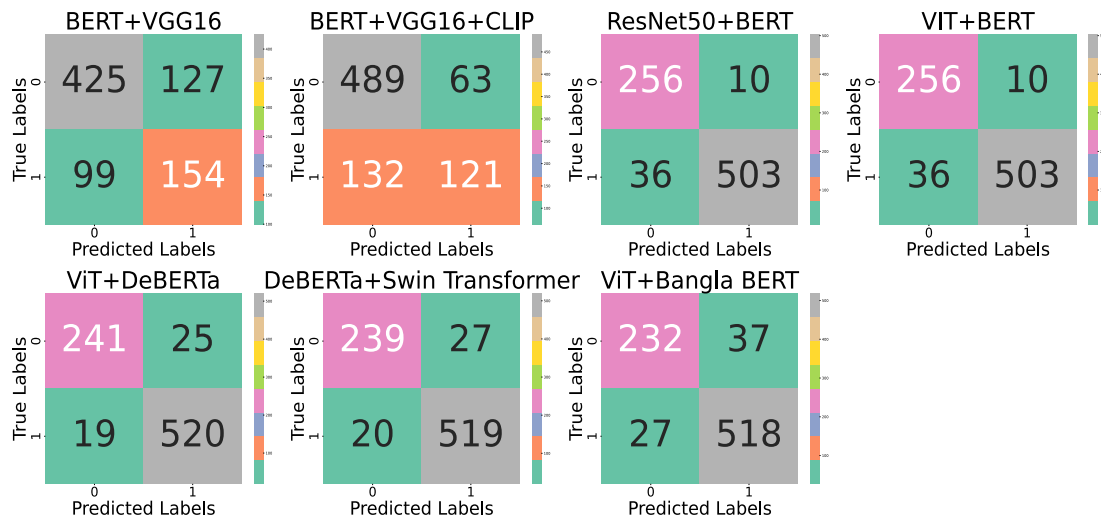


FIGURE 6. Multimodal LMs confusion matrix.

Memory usage was a critical factor in model selection and evaluation. The ViT + DeBERTa model utilized about 60% of the available GPU memory, while simpler models like VGG16 consumed around 40%. This highlights the trade-off between model complexity and resource utilization.

Inference speed was also measured to assess real-time application potential. The ViT + DeBERTa model had an average inference time of 0.15 seconds per image-text pair, making it suitable for real-time content moderation systems. Comparatively, GPT-based models took longer, with an inference time of approximately 0.25 seconds per pair, indicating a need for optimization in large language models when applied to real-time systems.

H. DISCUSSION

Integrating visual and textual modalities significantly enhances the effectiveness of troll meme detection. This study demonstrates that multimodal models, which leverage visual and textual information, outperform unimodal approaches. Among the evaluated models, the combination of ViT and DeBERTa achieved the highest accuracy of 94.287%, significantly surpassing other models. This result underscores the potential of transformers in handling multimodal tasks, as ViT effectively captures visual features while DeBERTa provides robust textual understanding. The superior performance of the MemesViTa model can be attributed to several factors. Firstly, transformers have shown exceptional capabilities in computer vision and NLP tasks. The self-attention mechanism in transformers allows these models to focus on relevant parts of the input data, whether an image or text, thus capturing intricate patterns and contextual relationships [35]. By integrating ViT and DeBERTa, the model benefits from the strengths of both architectures, leading to a more comprehensive analysis of memes [3], [13].

LSTM models underperformed compared to CNN and transformer-based models due to their focus on sequential data. Troll memes, however, require capturing both visual and textual patterns, where CNNs excel in extracting spatial features from images, and transformers like ViT and DeBERTa leverage self-attention mechanisms to capture global dependencies across modalities. This ability to handle the complex interplay between image and text gives transformers a significant advantage, leading to higher accuracy in troll meme detection.

The ViT model excels in extracting detailed visual features due to its transformer-based architecture, which processes image patches like tokens in text. This approach enables ViT to capture long-range dependencies and complex patterns in visual data [13]. On the other hand, DeBERTa's bidirectional transformer architecture is adept at understanding the context and semantics of textual data, making it highly effective for analyzing meme captions [36]. The synergy between ViT and DeBERTa allows the multimodal model to leverage visual and textual cues, resulting in superior troll meme detection performance.

In addition to the success of the MemesViTa model, the experiments also highlight the promise of few-shot learning with advanced LLMs like GPT-4. In our evaluation, GPT-4 achieved an accuracy of 80.00% in a 50-shot scenario, demonstrating its ability to generalize from a limited number of examples. This capability is precious in real-world applications where labeled data might be scarce or expensive [3]. Few-shot learning allows models to quickly adapt to new tasks with minimal training data, making it a practical solution for dynamic and evolving content such as troll memes.

The effectiveness of GPT-4 in few-shot learning scenarios can be attributed to its extensive pretraining on diverse datasets, which equips it with a broad understanding of

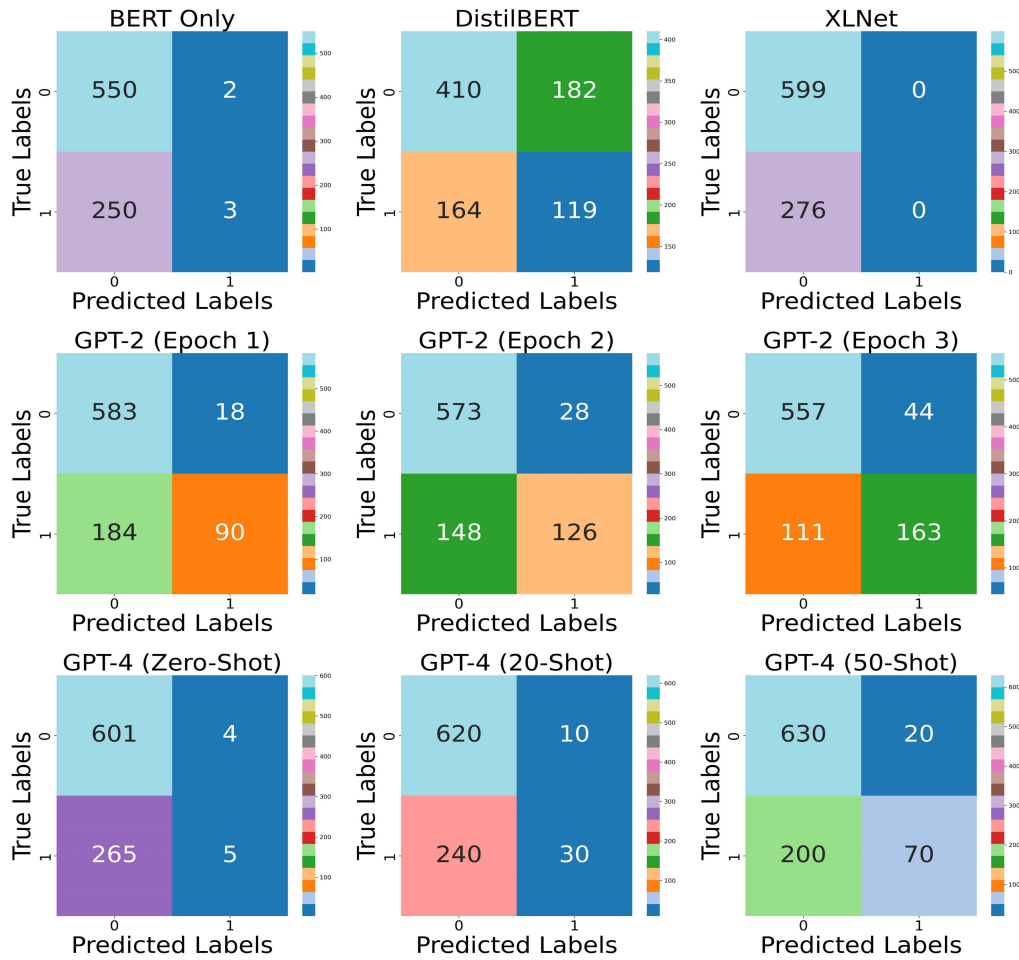


FIGURE 7. LMs and LLMs confusion matrix.

language and context. This extensive pretraining enables GPT-4 to recognize patterns and make accurate predictions even with limited task-specific data [3]. The results indicate that LLMs like GPT-4 can significantly enhance the capabilities of multimodal systems, providing a powerful tool for troll meme detection.

I. COMPARATIVE ANALYSIS OF STATE-OF-THE-ART METHODS

Table 7 provides a comprehensive overview of recent advancements in detecting and correcting hate speech in multimodal memes. The table summarizes various approaches, methodologies, dataset characteristics, classification strategies, and performance results from several notable studies in the field.

This research [16] demonstrated the effectiveness of zero-shot prompting with the LLaVA model, achieving an AUROC of up to 67.07% when OCR text was included. This work [17] employed a multimodal method incorporating image captioning, achieving an AUROC score of 78.86. This work [20] introduced a multimodal debate mechanism

with LLMs, achieving 87.00% accuracy. The GOAT-Bench study by [20] used models like GPT-4V, achieving 72.11% accuracy. This study [21] proposed the PromptHate model with RoBERTa, achieving an AUC of 90.96% and an accuracy of 84.47% on the Harm dataset. This study [18] focused on few-shot learning, achieving a VQA V2 zero-shot accuracy of 59.28% and a score of 60.33% on OKVQA. They [22] used parallel processing branches, achieving a 0.4298 macro F1 score on the Motion data. This research [23] developed the MOMENTA framework, achieving 84.17% accuracy on Harm-C and Harm-P datasets, respectively. Finally, the proposed model, named **MemesViTa**, showed superior performance with an accuracy of 94.287%. This analysis highlights significant advancements and varying methodologies in the field, providing a clear perspective on current research trends.

However, while the results are promising, there are several considerations and potential areas for improvement. First, the performance of zero-shot learning with GPT-4 was lower compared to few-shot scenarios. This suggests that while LLMs can generalize from minimal data, they benefit

TABLE 7. State of the art in detecting and correcting hate speech in multimodal memes.

Ref.	Models	Approach	Result
[16]	LLaVA (VLM)	Zero-shot prompting for hateful meme detection and correction.	AUROC of 65.77% and 63.92% without OCR text; 67.07% and 64.12% with OCR text.
[17]	Multimodal method incorporating image captioning	Integrating image captioning into hateful meme detection process.	AUROC score of 78.86.
[19]	Explainable harmful meme detection through multimodal debate	Using multimodal debate mechanism with LLMs.	Accuracy of 87.00%, macro averaged F1 score of 86.41% on Harm-C; 90.73% and 90.72% on Harm-P; 75.97% and 75.39% on FIIM.
[20]	GOAT-Bench	Benchmark for assessing hatefulness, misogyny, offensiveness, sarcasm and harmful content in memes.	Accuracy of 72.17%, macro-averaged F1 score of 70.29%.
[21]	PromptHate	Prompt-based model for hateful meme classification using pre-trained RoBERTa language model.	AUC of 90.96% and accuracy of 84.47% on Harm dataset; AUC of 81.45% and accuracy score of 72.98% on FIIM dataset.
[18]	Multimodal few-shot learning with frozen language models	Training vision encoder to represent images as continuous embeddings for few-shot learning.	VQA V2 zero-shot accuracy of 59.28%, score of 60.33% on OKVQA; 70.96% accuracy on ImageNet, 59.12% score with Visual Genome.
[22]	Multimodal deep neural networks	Two parallel processing branches (textual and visual) joined in a final classification step.	Macro F1 score of 0.2498, 8.24% improvement over baseline on Memotion dataset.
[23]	MOMENTA	Multimodal framework for detecting harmful memes and their targets.	Accuracy on Harm-C and Harm-P datasets: 77.95% and 78.54%, respectively; F1 scores: 69.65% and 68.83%.
Proposed model	MemesViTa	Joint image and text processing for multimodal hateful meme detection.	Accuracy: 94.287%, F1 Score: 95.82%, Precision: 95.82%, Recall: 96.42%.

significantly from task-specific examples [37]. Moreover, multimodal models' computational complexity and resource requirements, especially those involving transformers, can be substantial. Efficient training and inference strategies, such as model pruning or distillation, could be explored to make these models more accessible and feasible for deployment in real-world applications [38], [39].

Another area of interest is the interpretability of multimodal models. Understanding how these models make decisions and what features they prioritize can provide valuable insights, leading to more transparent and trustworthy systems [40], [41]. The MemesViTa model showcases the potential of transformers in handling complex multimodal tasks, while few-shot learning with GPT-4 demonstrates the practicality and effectiveness of advanced LLMs.

V. CONCLUSION AND FUTURE WORK

This study comprehensively evaluates multimodal deep-learning approaches for troll meme detection, addressing a critical issue in content moderation on social media platforms. By integrating visual and textual models, we demonstrated significant improvements in detecting troll memes, which often contain complex and nuanced information that single-modality models may overlook. Our experiments reveal that the ViT and DeBERTa combination outperforms other models, achieving the highest accuracy of 94.287%. This highlights the powerful synergy between advanced visual and textual transformers, effectively capturing and analyzing the multimodal nature of memes. Additionally,

applying few-shot learning with GPT-4 shows promising results, achieving an accuracy 80% in a 50-shot scenario, indicating that LLMs can generalize well from limited examples and adapt quickly to new tasks.

The models struggled with troll memes that used subtle sarcasm, ambiguous visuals, or complex text-image interactions. Misclassifications often occurred with memes where the visual and textual cues were hard to align or when cultural context was needed for accurate detection. These limitations suggest a need for better feature fusion and context-aware improvements in future models. Future research will employ advanced LLMs beyond GPT-4 to enhance NLP capabilities. Efforts will include expanding datasets to include diverse meme types and languages to improve model generalization. We will explore enhancements in zero-shot learning through advanced contextual information and domain adaptation. Additionally, we plan to develop efficient training and inference strategies, like model pruning and quantization, to reduce computational costs.

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